Frontier



Trent University and University of Alberta

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Trent University and University of Alberta

Organization location (we welcome applicants from anywhere in the world)

Ontario and Alberta, Canada

Name(s) of primary point(s) of contact for this application

Ian Power and Sasha Wilson

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

University

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Reactive phase targeting for mineral-based carbon removal

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Enhanced surficial mineralization has immense potential for carbon dioxide removal (CDR), yet large-scale deployment is constrained by a fundamental trade-off between reaction rates and costs. Most abundant alkaline feedstocks, such as ultramafic (Mg-rich) mine tailings, are mainly composed of recalcitrant silicate minerals, which react too slowly under ambient conditions. However, in some cases, these bulk feedstocks contain minor abundances of highly reactive phases that are diluted within the more recalcitrant material. Existing acceleration strategies often require energy-intensive grinding, heat or microwave activation, or chemical pretreatment, increasing capital and operating costs. Our project addresses this crucial challenge through a fundamentally different strategy: selectively extracting and concentrating highly reactive mineral phases, such as brucite [$Mg(OH)_2$], from ultramafic rock (**Fig. 1**), making it possible to focus carbonation only on this reactive fraction. Further, we aim to optimize carbonation efficiency by developing an empirical framework for predicting the rate of CO_2 drawdown into these minerals as a function of weather, climate or even the microclimate within a tailings storage facility.

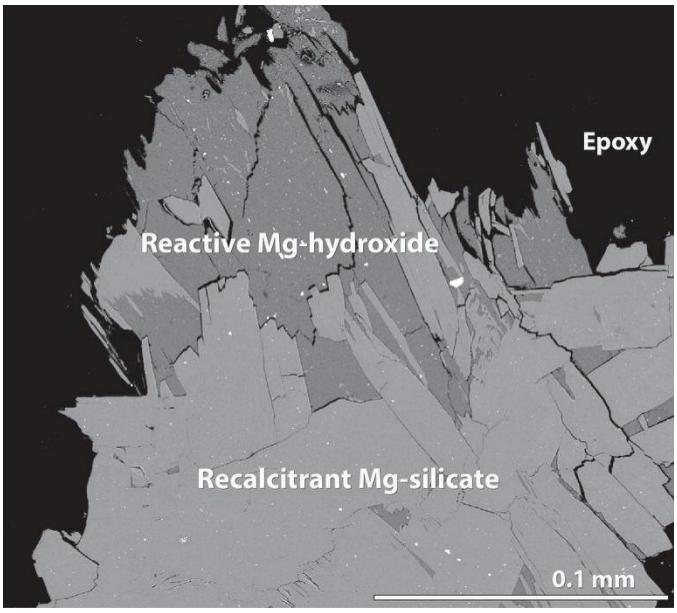


Figure 1. Image of ore from the Baptiste nickel deposit in Central British Columbia, Canada. The lighter grey mineral is serpentine (Mg-silicate), an abundant alkaline mineral that is suitable for CDR when pre-treated to enhance carbonation rates. The darker grey mineral is brucite (Mg-hydroxide), a highly reactive phase that typically occurs in serpentine-rich rocks at abundances of <1 to ~15 wt.%, which carbonates rapidly with atmospheric CO_2. Also note the bright spots that are the nickel-iron alloy (awaruite) that is intended to be extracted at this prospective mine. (Image modified from Power et al., 2020, International Journal of Greenhouse Gas Control)

This project will be jointly led by Prof. Ian Power, Canada Research Chair and Principal Investigator of the Power Geochemistry Laboratory at Trent University and Prof. Sasha Wilson, Canada Research Chair and Principal Investigator of the Environmental Economic Geology Laboratory at the University of Alberta. This project asks a decisive question for the mineral-based CDR field: **Can reactive-phase concentration improve ambient-air carbonation rates sufficiently to justify the cost and energy of separation?**

Our combined research programs provide the scientific basis for this approach. Together, we helped establish that brucite is the dominant driver of rapid carbonation in ultramafic mine tailings, beginning

with seminal research at the Mount Keith Nickel Mine in Australia (Wilson et al., 2014). Subsequent studies by our groups demonstrated that carbonation of periclase (MgO), lime (CaO) and portlandite [$Ca(OH)_2$] is limited not only by their abundance but also by CO_2 supply constraints, water availability, and mineral surface passivation under ambient conditions (Rausis et al., 2022; Dostie et al., 2024). Thus, optimizing carbonation pathways is crucial for developing surficial mineralization processes focused on these reactive phases.

This project will advance **Frontier's Enhancing Surficial Mineralization** priority area by focusing R&D activities on the most reactive minerals for direct air capture, including a low-energy feedstock processing approaches that increases effective carbonation efficiency. Rather than processing all material equally, our approach aims to concentrate the small fraction of minerals already capable of rapid carbonation and expose them to passive direct air contact under ambient conditions. To further accelerate CO_2 drawdown, we will assess a range of temperature and atmospheric conditions (e.g., relative humidity), generating predictive carbonation rate laws. In addition to brucite, hydrotalcite group minerals (layered double hydroxides) that are also common in ultramafic rocks offer another reactive carbonation pathway for surficial mineralization that will be explored in this project (**Fig. 2**).

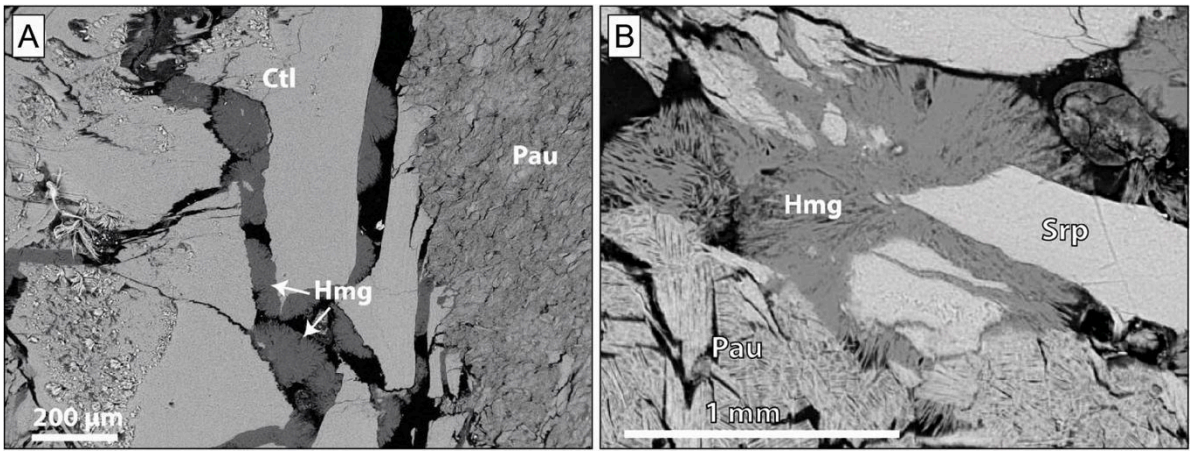


Figure 2. Images of mine wastes from Woodsreef Chrysotile Mine, New South Wales, Australia. (A) Hydromagnesite (Hmg) nucleation on chrysotile (Ctl), with pyroaurite (Pau). (B) Pyroaurite and hydromagnesite carbonate cement between disaggregated serpentine grains. These mineral associations strongly suggest that the hydrotalcite mineral pyroaurite plays a role in the carbonation of these wastes, forming hydrated Mg-carbonate minerals. (Images from Hamilton et al., 2018, International Journal of Greenhouse Gas Control)

We will test this concept using several representative feedstocks, including ultramafic tailings, waste rock, and ore (e.g., Baptist nickel deposit, Mount Keith nickel mine, and Thetford asbestos mine) that are selected to reflect real-world feedstock variability, and pure mineral samples for comparison (e.g., brucite ore and hydrotalcite hand specimens). Testing low-energy concentration methods (density separation, flotation, and leaching), we will quantify:

1. Recovery efficiency of reactive phases,
2. Ambient-air carbonation kinetics of enriched concentrates versus untreated bulk feedstocks versus pure mineral specimens.

The emphasis on **direct air capture under ambient conditions** is central to this proposal. If extracting and concentrating reactive phases is successful, this would validate a scalable pathway that

preserves the low-cost advantages of passive surficial mineralization while substantially increasing reaction rates.

The **broader impact on the CDR field** is potentially transformative. This project could unlock a scalable, lower-cost pathway for enhanced surficial mineralization by selectively concentrating the minerals that carbonate fastest under ambient conditions. If successful, it would improve carbon efficiency, reduce unnecessary processing of inert material, and accelerate deployment of mine-tailings-based durable CDR across diverse climates and feedstocks.

References

Dostie, L., Rausis, K., and Power, I.M., 2024, Passive direct air capture using calcium oxide powder: The importance of water vapor: Journal of Cleaner Production, 457:142394.

Power, I.M., Dipple, G.M., Bradshaw, P.M.D., and Harrison, A.L., 2020, Prospects for CO_2 mineralization and enhanced weathering of ultramafic mine tailings from the Baptiste nickel deposit in British Columbia, Canada: International Journal of Greenhouse Gas Control, 94:102895.

Rausis, K., Stubbs, A.R., Power, I.M., and Paulo, C., 2022, Rates of atmospheric CO_2 capture using magnesium oxide powder: International Journal of Greenhouse Gas Control, 119: 103701.

Wilson, S., Harrison, A.L., Dipple, G.M., Power, I.M., Barker, S.L.L., Mayer, K.U., Fallon, S.J., Raudsepp, M., and Southam, G., 2014. Offsetting of CO_2 emissions by air capture in mine tailings at the Mount Keith Nickel Mine, Western Australia: Rates, controls and prospects for carbon neutral mining. International Journal of Greenhouse Gas Control, 25: 121-140.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Project goals

This project is designed to determine whether selective extraction and concentration of highly reactive mineral phases can create a scalable, low-cost pathway for enhanced surficial mineralization. The overarching goal is to quantify whether reactive phase enrichment delivers sufficient carbonation gains to justify separation. In parallel, we must answer what environmental conditions and pathways are optimal for the carbonation of these reactive phases by quantifying fundamental dissolution and carbonate rates at the mineral-water-CO_2 interface.

The project will be led by Prof. Ian Power (Trent University) and Prof. Sasha Wilson (University of Alberta) whose groups bring together complementary expertise in surficial mineralization, carbonation kinetics, feedstock characterization, and experimental process development. Our collaboration integrates two internationally recognized research programs that have helped establish the scientific foundation for a variety of mineral-based CDR approaches, including CO_2 mineralization within mine wastes with enhanced metal recovery, microbial and enzyme-mediated carbonation, and, most relevant to this study, the carbonation of highly reactive phases, as well as the growth and

stability of carbonate minerals at Earth's surface conditions.

Research questions

This project is organized around two crucial research questions. First, can highly reactive phases be efficiently concentrated from mineralogically diverse feedstocks using low-energy methods that remain economically viable at scale? Reactive phases such as brucite tend to dominate carbonation of their bulk host rocks (e.g., ultramafic tailings) yet typically occur in minor amounts disseminated throughout abundant yet relatively recalcitrant silicates (e.g., serpentine). Therefore, the first task is to determine whether practical separation methods can recover reactive fractions at sufficient purity and yield. Second, does reactive phase enrichment materially improve carbonation rates under ambient atmospheric CO_2 conditions? Thus, a deep understanding of fundamental dissolution and carbonation processes is essential to optimize carbonation of reactive concentrates compared to untreated bulk feedstocks is essential.

Approaches and methods

Complete **feedstock characterization** will precede extraction and carbonation testing. Particle size distributions and specific surface areas will be determined using a laser scattering particle size analyzer and the Brunauer, Emmett and Teller (BET) method, respectively. Quantitative mineralogical data will be obtained by Rietveld refinement using powder X-ray diffraction data and using X-ray mapping techniques such as electron probe microanalysis. Geochemical data will be obtained by X-ray fluorescence (XRF) spectroscopy and inductively coupled plasma mass spectroscopy (ICP-MS) after digesting samples. The total inorganic carbon (TIC) content of the residues will be measured using coulometry. The majority of these methods are described by Stubbs et al. (2023) and Zeyen et al. (2022).

A variety of separation methods will be attempted to selectively extract reactive phases, such as brucite, from bulk feedstocks.

For this study, a **flotation method** adapted from Sun et al. (2012) will be performed to produce brucite-rich concentrates from ultramafic feedstocks, where brucite recovery exceeded 90% and serpentine recovery remained below 10%. This previous study demonstrated that brucite can be effectively separated from serpentine using sodium oleate as a collector and sodium silicate as a selective serpentine depressant under alkaline conditions. Representative ultramafic tailings will be subjected to flotation tests across controlled pH and reagent conditions to optimize brucite recovery.

Given that hydrotalcite group minerals (~2.0–2.2 g/cm^3) and brucite (~2.4 g/cm^3) are significantly less dense than their ultramafic host minerals, including serpentine (~2.5–2.7 g/cm^3), **density-based separation** may provide a low-energy pathway for reactive-phase enrichment. To test this approach, heavy liquid separation and centrifugal density classification will be used to isolate low-density fractions enriched in brucite and hydrotalcites. Flotation and density separation are also applicable to the mineral processing facilities at active mines.

Column leaching experiments will be conducted on representative ultramafic feedstocks using simulated rainwater, dilute inorganic acids, and weak organic acids to exploit the faster dissolution rates of brucite and hydrotalcite group minerals relative to serpentine (five orders of magnitude faster). Time-series leachates will be analyzed for Mg, Si, Fe, and pH to identify conditions that maximize Mg release. Residual solids will be reanalyzed to determine the efficiency of brucite and hydrotalcite group removal, and pure brucite will be precipitated from recovered Mg-rich leachates. This approach is applicable to historic mine wastes (e.g., Thetford asbestos tailings) and connects to

proposed heap leaching technologies.

The resulting concentrates will be characterized by powder X-ray diffraction to quantify reactive phase enrichment. The highest-purity fractions will be used in subsequent ambient-air carbonation experiments, comparing to bulk feedstocks and pure mineral specimens.

We will **measure CO_2 uptake by mineral sorbents** as a function of temperature, relative humidity, partial pressure of CO_2 and time. The resulting data will be used to determine temperature- and humidity-dependent rates of CO_2 uptake by brucite and hydrotalcite group minerals which are the key sorbents for CO_2 removal at critical metal mines. These data will allow us to establish temperature-dependent rate laws for CO_2 removal across all mineral-based technologies, which will improve the accuracy of TEA and LCA models.

We will use a combination of in situ powder X-ray diffraction and direct CO_2 flux measurements to determine carbonation rates for brucite and hydrotalcite group minerals in humidified air. Recent results published by Leila et al. (2025) have shown that hydrotalcite group minerals can be converted to nesquehonite ($MgCO_3 \cdot 3H_2O$), dramatically increasing their utility for CO_2 removal. Brucite and the most common hydrotalcite group minerals in ultramafic rocks, pyroaurite [$Mg_6Fe_2CO_3(OH)_{16}\cdot 4H_2O$], iowaite [$Mg_6Fe_2Cl_2(OH)_{16}\cdot 4H_2O$] and stichtite [$Mg_6Cr_2(CO_3)(OH)_{16}\cdot 4H_2O$], will be investigated under relative humidities between 0% and 100% and temperatures from -20 °C to +60 °C. A dynamic water vapour sorption analyzer will be used to measure the thickness of water films at mineral surfaces to determine the threshold thickness required to initiate carbonation reactions. Scanning electron microscopy and total inorganic carbon results will be correlated with X-ray diffraction data to map the carbonation efficiency in temperature–humidity space based on the extent of passivation.

Project location and scale

The proposed work will be carried out across two established research groups with complementary capabilities. At Trent University, Ian Power's group will primarily conduct the reactive phrase extraction and concentration experiments, feedstock characterization, and assist with carbonation kinetics experiments by completing inorganic carbon quantification. At the University of Alberta, Sasha Wilson's group will lead carbonation kinetics experiments, contribute to feedstock characterization and separation optimization. The project is designed at the laboratory bench scale, but separation techniques being tested are directly transferable to mining and mineral processing, and the carbonation experiments are relevant to real-world passive surficial CO_2 mineralization.

Best-in-class approach

Our approach distinguishes itself from other surficial mineralization strategies by addressing the mineralogical bottleneck directly rather than attempting to force faster carbonation through energy-intensive inputs. Recovery methods were selected because they are scalable, industrially familiar, and comparatively low in energy demand compared to technologies being pursued for accelerating carbonation of recalcitrant silicate minerals.

Reactive phase targeting is fundamentally different because it maximizes the natural reactivity already present in the feedstock. Rather than attempting to react all material, including recalcitrant phases, we aim to isolate the fraction that is already capable of rapid surficial mineralization. This preserves the low-CapEx and low-OpEx advantages of surficial mineralization while potentially achieving much higher reaction rates per tonne processed. Furthermore, by determining fundamental dissolution and carbonation rates at mineral surfaces, we will be able to optimize carbonation across a wide range of

CDR applications.

References not previously listed

Leila, M., Hazlett, R., George, P. M., Šegvić, B., and Fustic, M., 2024, Low temperature carbonation and CO_2 mineral trapping in altered hydrotalcite-rich ultramafic rocks: Scientific Reports, 15:31133.

Stubbs, A.R., Power, I.M., Paulo, C., Wang, B.L., Zeyen, N., Wilson, S., Mervine, E., and Gunning, C., 2023, Impact of wet-dry cycles on enhanced rock weathering of brucite, wollastonite, serpentinite and kimberlite: Implications for carbon verification: Chemical Geology, 637:17.

Sun, D.Y., Yin, W.Z., Zhu, D.S., Liu, M.B., and Luo, X. M., 2012, Flotation characteristics of brucite and serpentine: Advanced Materials Research, 454:199-204.

Zeyen, N., Wang, B., Wilson, S., Paulo, C., Stubbs, A. R., Power, I. M., Steele-MacInnis, M., Lanzirotti, A., Newville, M., Paterson, D. J., Hamilton, J. L., Jones, T. R., Turvey, C. C., Dipple, G. M., and Southam, G., 2022, Cation exchange in smectites as a new approach to mineral carbonation: Frontiers in Climate, 4:913632.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

The primary **scientific and technical risk** lies in the high-risk, high-reward component (***breakthrough "wildcard"***) of the project: whether reactive phases, such as brucite and hydrotalcite group minerals, can be sufficiently concentrated through extraction and separation to increase ambient-air carbonation and in an economical manner that justifies separation costs. To reduce this uncertainty, we will evaluate three separation approaches: flotation, density separation, and leaching using at least three representative ultramafic materials. This comparative design ensures that if one method underperforms, alternative pathways remain available. Even if performance gains are modest, the project will still provide a decisive result for the CDR field.

In contrast, the determination of fundamental dissolution and carbonation rates across temperature and humidity conditions is **low risk and guaranteed to deliver** valuable, broadly applicable datasets. These experiments build directly on established methods and will produce predictive rate laws applicable across mineral-based CDR pathways, regardless of separation outcomes. Furthermore, the CO_2 reactivity of mineral sorbents has not previously been explored at sub-zero temperatures. However, a range of temperatures relevant to surficial mineralization will be tested. If sub-zero carbonation proves too slow for direct kinetic quantification, these conditions will still provide a lower constraint for predictive rate-law models.

Project execution risk is low as our laboratories and universities already possess the required facilities, including mineral processing equipment, carbonation chambers, and analytical instrumentation. Our groups have worked extensively together to execute a wide variety of CDR projects, particularly CO_2 sequestration within mine wastes containing reactive phases. Coordination will be managed through the milestones and regular team meetings.

MRV risk is negligible with the carbonation of highly reactive phases. To ensure robust measurement,

carbonation extents will be verified by total inorganic carbon analysis by coulometry (detection range of 1–10,000 µg C and precision of 0.01 µg C), and mineralogical transformations will be monitored by quantitative X-ray diffraction.

Financial and ecosystem risks are also very low. This is a laboratory-based project and therefore, poses no risk to ecosystems. We have access to feedstock materials through our archived sample collections from prior studies and existing partnerships, minimizing supply uncertainty. Because the project relies primarily on existing facilities, budget risk is limited.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	1. Validated reactive phase separation datasets quantifying recovery efficiency across a variety of separation techniques 2. Comparative carbonation performance datasets measuring rates and extents for enriched concentrates, untreated bulk feedstocks, and pure reactive mineral phases under controlled temperature and humidity conditions, enabling prediction of CO_2 removal performance across deployment environments
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	This project will determine whether extracting and concentrating reactive minerals can materially increase carbonation efficiency under ambient conditions, clarifying a major unanswered question in surficial mineralization CDR. It will provide predictive kinetic tools to improve confidence in deployment planning across different climates and feedstocks.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	**Project duration: 24 months** **Months 1–3:** Additional feedstocks will be acquired and combined with archived samples. Baseline

	mineralogical and geochemical characterization will be completed. **Months 4–20:** Trent University will lead flotation, density separation, and selective leaching to optimize recovery of brucite- and hydrotalcite-rich fractions, while in parallel the University of Alberta will conduct carbonation kinetics experiments on enriched concentrates, bulk feedstocks, and pure mineral phases under controlled temperature and humidity conditions. **Months 21–24:** Both teams will integrate datasets, finalize validated separation and carbonation performance results, and prepare deliverables (e.g., publications and reports).
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	US$385k